A Degree-Independent Lorentzian Classification with Prescribed Variable Cutoffs

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Results obtained: October 2026
Manuscript: 22 October 2026 · Version 1.0.0

Abstract

We classify Lorentzian membership in a two-parameter family of factorial-normalized symmetric polynomials whose coefficients depend on support size and on the numbers of singleton and doubleton exponents. For every total degree at least six, the same explicit parameter criterion applies: apart from a normalized Schur boundary point, membership is controlled by one negative-semidefinite matrix of order two depending only on the number of variables. The proof reduces every derivative Hessian to four occupancy classes, treats exceptional profiles separately, and establishes the remaining signs through explicit positive polynomial identities. A specialization gives a sharp weighted Giambelli criterion, including the normalized permanent for every shape $(m, 3)$ with $m \geq 3$. Another specialization realizes every prescribed variable cutoff using rational parameters that work simultaneously in all degrees at least six. This cutoff phenomenon is already known in other families; the contribution here is its exact realization inside the classified family and its independence of total degree. The general Giambelli-immanant conjecture remains open.

Contribution and significance

Known Lorentzian criteria turn the problem into a support condition and inertia tests for derivative Hessians. Symmetric block compression is also established. The difficulty in the present family is to show that one small controlling derivative suffices for every other derivative profile, every variable count, and every degree at least six. The proof supplies this reduction, including singular parameter boundaries, through explicit congruences and positive coefficient identities rather than finite-degree extrapolation.

The classification has two concrete consequences. On a weighted Giambelli line, it determines the entire allowed weight range, not merely a positive example. On another rational line, it gives simultaneous sharp variable cutoffs in every admissible degree. General cutoff examples already follow from cubic criteria, so that phenomenon is not claimed as new. The application is a restricted rank-two family; it neither proves a theorem for arbitrary Giambelli immanants nor derives Lorentzianity from an unjustified closure under positive linear combinations.

1. Introduction and conventions

We give a degree-independent Lorentzian classification for a two-parameter symmetric coefficient family. Its coefficients depend only on support size and the numbers of exponents equal to one and two. A special line recovers a weighted rank-two Giambelli expression; the classification also gives rational examples with any prescribed variable-count cutoff, simultaneously in every degree at least six.

We use factorial normalization $\mathcal{N}(\sum_\gamma c_\gamma x^\gamma) = \sum_\gamma c_\gamma x^\gamma/\gamma!$, where $\gamma! = \prod_i \gamma_i!$. For a nonnegative integer vector γ, let $\ell(\gamma)$ be its number of positive entries, and let $u(\gamma)$ and $v(\gamma)$ be the numbers of entries equal to one and two. The zero polynomial is included in the Lorentzian class.

A homogeneous polynomial with nonnegative coefficients is Lorentzian if and only if its support is M-convex and every derivative of degree two has Hessian with at most one positive eigenvalue [1, Theorem 2.25]. Here J is M-convex if, for $\alpha, \beta \in J$ and $\alpha_i > \beta_i$, some j with $\alpha_j < \beta_j$ has both $\alpha - e_i + e_j, \beta + e_i - e_j \in J$. No strictness or nondegeneracy is assumed.

Symmetric Hessian compression and limiting zero-coordinate matrices are established methods [3, Section 3.3]. The contribution here is the explicit parameter region and its independence of degree for this family, rather than a new general compression method. The weighted application supplies a restricted case associated with Zhang's Giambelli-immanant conjecture [4, Conjecture 5.4]; it does not settle general rank-two immanants.

2. Definition and classification

All parameters are real. Let $N \geq 6$ and $n \geq 1$. Define

$$F_{N,n}^{p,s}(x) = \sum_{\substack{\gamma \in \mathbb{N}^n \\ |\gamma| = N}} c_\gamma \frac{x^\gamma}{\gamma!}, \qquad c_\gamma = p\binom{\ell(\gamma)+1}{3} - s(\ell(\gamma)-1)u(\gamma) - v(\gamma).$$

Write

$$a = p - s, \quad w = 3p + 1 - 4s, \quad L = 2p - ps - 1, \quad T = 7s - 6p, \quad \delta = 6p - 7s + 1,$$

and

$$\mathcal{D} = LT - a(3 - 2s)^2.$$

When $a > 0$ and $q > 0$, put

$$S_q = \begin{pmatrix} L/a & 3 - 2s \\ 3 - 2s & T - \delta/q \end{pmatrix}, \qquad S_\infty = \begin{pmatrix} L/a & 3 - 2s \\ 3 - 2s & T \end{pmatrix}.$$

Theorem 2.1. *Fix $N \geq 6$. For $n = 1$, $F_{N,1}^{p,s} = 0$ for every p, s. For $n = 2$, the polynomial is Lorentzian if and only if*

$$p \geq 1, \qquad s \leq p, \qquad L \leq 0.$$

For $n \geq 3$, it is Lorentzian if and only if either $(p, s) = (1, 1)$, or

$$p > s > 1, \qquad w \geq 0, \qquad S_{n-2} \leq 0.$$

The last matrix condition is equivalent to

$$L \leq 0, \qquad T - \frac{\delta}{n-2} \leq 0, \qquad \mathcal{D} - \frac{L\delta}{n-2} \geq 0.$$

Corollary 2.2. *For each fixed $N \geq 6$, the polynomial is Lorentzian for every finite variable count exactly when $(p, s) = (1, 1)$, or*

$$p > s > 1, \quad 3p + 1 \geq 4s, \quad L \leq 0, \quad T \leq 0, \quad \mathcal{D} \geq 0.$$

Moreover,

$$\mathcal{D} = 6p^2(s - 2) - p(11s^2 - 26s + 3) + 4s^3 - 12s^2 + 2s.$$

3. Necessary conditions

Factorial normalization yields

$$(\nabla^2 \partial^\beta F^{p,s}_{N,n})_{ij} = c_{\beta+e_i+e_j}, \qquad |\beta| = N-2.$$

The contents $(N-1,1)$ and $(N-2,2)$ give $a \geq 0$ and $p \geq 1$.

For $\beta = (N-2, 0, \ldots, 0)$, the submatrix on the zero coordinates has diagonal $p-1$ and off-diagonal $4a$. Their difference is $-w$. If $n \geq 3$ and $w < 0$, this submatrix is positive definite: its contrast eigenvalues are $-w > 0$, and its remaining eigenvalue is $p - 1 + 4a(n-2) > 0$. Thus $w \geq 0$. If $a = 0$, then $w = 1 - p$, forcing $p = s = 1$.

Suppose $a > 0$. The derivative profile $(N-3, 1)$ gives the block

$$\begin{pmatrix} a & p-1 \\ p-1 & p \end{pmatrix},$$

with determinant L. Hence $L \leq 0$, implying $s \geq 2 - 1/p \geq 1$. Equality $s = 1$ would imply $p \leq 1$, contrary to $p > s$, so $p > s > 1$.

For $q = n - 2$, the class-average matrix of this derivative is

$$R_q = \begin{pmatrix} a & p-1 & 4a \\ p-1 & p & 4p-2s-1 \\ 4a & 4p-2s-1 & 10p-9s-\delta/q \end{pmatrix}.$$

The Schur complement of its positive pivot a is S_q. Thus at most one positive eigenvalue requires $S_q \leq 0$.

4. Support and class reduction

Throughout this section assume

$$p > s > 1, \qquad w \geq 0. \tag{A}$$

Every mixed coefficient is positive. For support size $r = 2$, its possible values are $a, p-1, p$. For $r = 3$, two singleton entries force the remaining exponent to be at least four, giving $4a$; otherwise the coefficient is at least $4p - 2s - 2 > 0$. For $r = 4$, the all-singleton content is impossible, and

$$c_\gamma \geq 10p - 9s - 1 = 10a + s - 1 > 0.$$

For $r \geq 5$, using $v \leq r - u$ and $s(r-1) > 1$ gives

$$c_\gamma \geq r(r-1)\left(\frac{p(r+1)}{6} - s\right) > 0.$$

Pure powers have coefficient zero. Consequently,

$$\operatorname{supp} F^{p,s}_{N,n} = \{\gamma \geq 0 : |\gamma| = N,\ \gamma_i \leq N-1\}.$$

This set is M-convex. If $\gamma_i > \eta_i$, choose j with $\gamma_j < \eta_j$. Both exchanged compositions remain nonnegative and satisfy the upper bounds.

Partition derivative coordinates into

$$H : \beta_i \geq 3, \quad V : \beta_i = 2, \quad U : \beta_i = 1, \quad Z : \beta_i = 0.$$

Let their sizes be h, v, u, q and let $r = h + v + u$. Set

$$\begin{aligned} K &= p\binom{r+1}{3} - s(r-1)u - v, & J &= K + v, \\ A &= s(r-1) - 1, & d &= A - 1, \\ E &= p(r+1) - 2s, & B &= rE/2 - su, \\ C &= sr - 1, & \Delta &= B + E - C. \end{aligned}$$

Coefficient differences give the within-class contrast eigenvalues $0, -1, -d, -\Delta$. The limiting class-average matrix, retaining an algebraic heavy row, is

$$M_\infty = \begin{pmatrix} K & K+1 & K+A & K+B \\ K+1 & K+2-1/v & K+A+1 & K+B+1 \\ K+A & K+A+1 & K+2A-d/u & K+B+C \\ K+B & K+B+1 & K+B+C & K+2B+E \end{pmatrix}. \tag{1}$$

Delete absent V, U classes before evaluating their denominators. Delete the artificial heavy row when $h = 0$, and delete Z when $q = 0$. If $q > 0$, subtract Δ/q from the final diagonal entry.

All occurring contrast eigenvalues are nonpositive under (A). For $r \geq 3$, write $x = r-3$, $y = s-1$. At $u = r$,

$$6\Delta = 3x^2 + 9x + y(4x^2 + 18x + 14) + w(x^2 + 9x + 20) > 0.$$

For smaller u, add $s(r-u)$ to Δ. For $r = 2$, its values are $6a + 1$ or $y + 2w$. For $r = 1$, it is w. The possibly negative d when $r = 2, u = 1$ produces no contrast, because the singleton class has size one.

5. Generic inertia proof

For $r \geq 2$, $K > 0$ except for the controlling profile $(N-3, 1)$ and the three profiles

$$(2,2), \qquad (2,1,1), \qquad (1,1,1,1). \tag{2}$$

Indeed, for $r \geq 5$,

$$K \geq r(r-1)\left(\frac{p(r+1)}{6} - s\right) > 0.$$

For $r = 4, u \leq 3$, $K \geq 10a + s - 1 > 0$. For $r = 3$, only $u = 2, v = 1, h = 0$ is exceptional: when $u \leq 1$, $K \geq 4p - 2s - 2 > 0$, and when $u = 2, h = 1$, $K = 4a$. For $r = 2, u = 0$, only $v = 2, h = 0$ can have nonpositive K.

Pivoting on $K > 0$ and then, if present, on V gives the negative pivot

$$-\frac{1}{v} - \frac{1}{K}$$

and the remaining U, Z matrix

$$S = \begin{pmatrix} -d/u - A^2/J & s - AB/J \\ s - AB/J & E - B^2/J \end{pmatrix}. \tag{3}$$

Absent classes are deleted.

Put

$$J_0 = p\binom{r+1}{3}, \qquad B_0 = rE/2, \qquad Z_0 = B_0^2 - EJ_0.$$

Then

$$Z_0 = \frac{rE}{12}\,[p(r+1)(r+2) - 6sr] > 0$$

for $r \geq 2$: $E > 0$, and the bracket is positive since $p > s > 1$. If $u = 0$, this proves $E - B_0^2/J_0 < 0$.

For $u > 0, r \geq 3$, define

$$D = d(B^2 - EJ) - u(EA^2 - 2sAB + s^2J).$$

Direct expansion gives

$$\det S = \frac{D}{uJ}, \tag{4}$$

and

$$D = dZ_0 + u(2sB_0 - E - s^2J_0). \tag{5}$$

Thus D is affine in u. Its $u = 0$ endpoint is positive.

For $r = 3$, admissibility gives $u \leq 2$, and

$$9D(3,2) = 24w + 6y + 96yw + 40yw^2 + 24y^2 + 128y^2w + 34y^3 > 0. \tag{6}$$

For $r \geq 4$, set $x = r - 4$. We have

$$108D(r,r) = r \sum_{i,j} y^i w^j P_{ij}(x), \tag{7}$$

where all coefficients in the following table are positive. Entries are in ascending powers of x.

(i,j)	1	x	x^2	x^3	x^4
$(0,0)$	54	153	153	63	9
$(0,1)$	540	648	318	72	6
$(0,2)$	150	235	101	17	1
$(1,0)$	378	945	813	279	33
$(1,1)$	1740	2078	946	190	14
$(1,2)$	450	405	133	19	1
$(2,0)$	780	1870	1376	398	40
$(2,1)$	1620	1734	692	122	8
$(3,0)$	576	1212	748	184	16

Equations (5)–(7) imply $D > 0$ throughout the admissible interval of u. Since the first diagonal of (3) is negative, (4) makes S negative definite. Thus the generic augmented matrix has exactly one positive eigenvalue. Principal restriction and subtraction of $\Delta/q \geq 0$ cannot increase that index.

6. Exceptional profiles and completion

For one heavy coordinate, the nontrivial class matrix is

$$\begin{pmatrix} 0 & a \\ a & 4a - w/q \end{pmatrix},$$

with determinant $-a^2 < 0$ and nonpositive contrast directions. The profile $(N-3, 1)$ is governed exactly by S_q.

For $(2,2)$ the limiting matrix is

$$\begin{pmatrix} p - \frac{1}{2} & 4p - 2s - 1 \\ 4p - 2s - 1 & 10p - 6s - 2 \end{pmatrix},$$

whose determinant is $-a(6p - 4s - 1) < 0$.

For $(2, 1, 1)$ the limiting matrix, in V, U, Z order, is

$$R = \begin{pmatrix} 4a & 4a + 2s - 1 & 10p - 9s \\ 4a + 2s - 1 & 4a + 3s - 2 & 10p - 6s - 2 \\ 10p - 9s & 10p - 6s - 2 & 20p - 16s - 1 \end{pmatrix}.$$

Its V pivot is positive, and its V, U principal minor is

$$-4as - (2s - 1)^2 < 0.$$

Furthermore,

$$9 \det R = 12w + 20w^2 + 3y + 64yw + 20yw^2 + 17y^2 + 64y^2w + 17y^3 > 0.$$

The residual after the positive pivot is therefore negative definite.

For $(1, 1, 1, 1)$ the limiting matrix is

$$R = \begin{pmatrix} 10p - \frac{27}{4}s - \frac{3}{2} & 20p - 16s - 1 \\ 20p - 16s - 1 & 35p - 30s \end{pmatrix}.$$

Its first diagonal is positive under (A), and

$$36(-\det R) = 9 + 105w + 200w^2 + 69y + 355yw + 146y^2 > 0.$$

All finite zero-class corrections are nonpositive. These arguments prove sufficiency for $n \geq 3$.

At $(p, s) = (1, 1)$,

$$\sum_{|\gamma|=N} c_\gamma x^\gamma = h_{N-3}h_3 - h_{N-2}h_2 = s_{(N-3,3)}.$$

To verify the coefficients, capped compositions of three and two give respectively

$$\binom{r+2}{3} - ru - v, \qquad \binom{r+1}{2} - u.$$

Their difference is the stated c_γ. Normalized Schur Lorentzianity follows from [2, Theorem 3].

In two variables, factorial normalization turns the relevant ultra-log-concavity condition into ordinary log-concavity of

$$0,\ a,\ p - 1,\ p, \ldots, p,\ p - 1,\ a,\ 0.$$

Nonnegativity, no internal zeros, and log-concavity are equivalent to

$$p \geq 1, \qquad s \leq p, \qquad (p - 1)^2 \geq p(p - s).$$

The last inequality is $L \leq 0$. For $p = 1$, these inequalities force $a = 0$, so there is no hidden internal-zero exception. In one variable all coefficients vanish. This completes the proof of Theorem 2.1.

Under (A), $\delta = (s - 1) + 2w > 0$ and

$$S_q = S_\infty - \operatorname{diag}(0, \delta/q).$$

Negative semidefiniteness is closed under limits and preserved by this subtraction. This proves Corollary 2.2.

7. Prescribed variable-count cutoffs in this family

Corollary 7.1. *For each integer $M \geq 2$, put*

$$p_M = \frac{M+2}{3}, \qquad s_M = \frac{M+3}{4}.$$

Then, for every integer $N \geq 6$ and every finite $n \geq 1$,

$$F_{N,n}^{p_M,s_M} \text{ is Lorentzian} \quad \Longleftrightarrow \quad n \leq M.$$

The same parameters work simultaneously for every $N \geq 6$. The coefficients d_γ in $12F_{N,n}^{p_M,s_M} = \sum_\gamma d_\gamma x^\gamma/\gamma!$ are

$$4(M+2)\binom{\ell(\gamma)+1}{3} - 3(M+3)(\ell(\gamma)-1)u(\gamma) - 12v(\gamma),$$

and are nonnegative integers.

Proof. The parameters satisfy

$$a = \frac{M-1}{12} > 0, \quad w = 0, \quad L = -\frac{(M-1)(M-2)}{12}, \quad T = \frac{5-M}{4}, \quad \delta = \frac{M-1}{4}.$$

Thus the one- and two-variable cases follow from Theorem 2.1. For $n \geq 3$ write $q = n-2$. The controlling matrix becomes

$$S_q = \begin{pmatrix} -(M-2) & (3-M)/2 \\ (3-M)/2 & (5-M)/4 - (M-1)/(4q) \end{pmatrix}, \quad \det S_q = \frac{(M-1)(M-2-q)}{4q}.$$

If $M \geq 3$, its first diagonal is negative, so it is negative semidefinite exactly when $q \leq M-2$. If $M = 2$, its determinant is $-1/4$ for every $q > 0$, and no $n \geq 3$ is allowed. Coefficient nonnegativity follows from the support proof under (A); multiplication by twelve clears all coefficient denominators. □

At the first failing variable count $n = M+1$, the residual determinant is always $-1/4$. For the derivative $\beta = (N-3, 1, 0^{n-2})$, the full Hessian has inertia $(2, M-1, 0)$ at this first failure. At $n = M$, it has inertia $(1, M-2, 1)$, with the binary case $M = 2$ obtained directly from its 2×2 block. These inertia statements follow from the positive pivot a, the residual matrix, and the negative zero-class contrasts $-\delta$.

The phenomenon of a prescribed variable cutoff is not new in general. For example, [3, Theorem 4.3] already implies it for cubics: for $M \geq 3$, coefficients $a = 1$, $b = 2$, $c = 4 + 2/(M-2)$ in the normalized monomial basis give a cutoff at M variables. Thus Corollary 7.1 concerns the displayed Giambelli-derived family and its simultaneous uniformity in every degree $N \geq 6$; it is not a claim to discover the general phenomenon.

8. A sharp weighted Giambelli line

Write h_j for the complete symmetric polynomial and s_λ for the Schur polynomial, always in the same n variables.

Corollary 8.1. *For every integer $m \geq 3$, every $n \geq 2$, and every real t,*

$$\mathcal{N}\big(s_{(m,3)} + t h_m s_{(2,1)}\big) \quad \text{is Lorentzian} \quad \Longleftrightarrow \quad t = 0 \text{ or } t \geq \tfrac{1}{2}.$$

For $n = 1$ this polynomial is zero for every t. In particular, the factorial-normalized Giambelli permanent for shape $(m, 3)$ is Lorentzian.

Proof. Set $N = m + 3$ and let r, u, v denote the three occupancy statistics. The Giambelli identity gives

$$s_{(m,3)} = s_{(m,1)}h_2 - h_m s_{(2,1)}.$$

The coefficient of x^γ in the two products on the right is, respectively,

$$3\binom{r+1}{3} - 2(r-1)u - v, \qquad 2\binom{r+1}{3} - (r-1)u.$$

For the second identity, $s_{(2,1)}$ has coefficient one on content $(2,1)$ and coefficient two on $(1,1,1)$, so direct counting gives $(r-u)(r-1) + 2\binom{r}{3}$. For the first identity, the coefficient of $s_{(m,1)}$ is the residual support size minus one. There are $r-u$ removable square contents and $\binom{r}{2}$ removable squarefree contents. Removing a square reduces the support exactly for the v doubletons; removing a squarefree content reduces it a total of $u(r-1)$ times. The resulting coefficient is $(r-1)(r-u+\binom{r}{2}) - v - u(r-1)$, as claimed. Consequently the normalized weighted expression is $F_{N,n}^{1+2t,1+t}$.

On this line,

$$L = t(1-2t), \quad T = 1-5t, \quad \mathcal{D} = 3t^2(2t-1), \quad w = 2t.$$

Theorem 2.1 proves sufficiency for $t \geq 1/2$ in every n, and the exceptional Schur point gives $t = 0$. For $t < 0$ the coefficient at $(N-1,1)$ is negative. For $0 < t < 1/2$, the two positive derivative coordinates in $\beta = (N-3,1,0,\ldots,0)$ give

$$\begin{pmatrix} t & 2t \\ 2t & 1+2t \end{pmatrix}, \qquad \det = t(1-2t) > 0,$$

a positive definite principal block. For the permanent, Giambelli's 2×2 matrix is $\left(\begin{smallmatrix} s_{(m,1)} & h_m \\ s_{(2,1)} & h_2 \end{smallmatrix}\right)$, so its permanent equals $s_{(m,3)} + 2h_m s_{(2,1)}$. □

9. Consequences and scope

For every fixed variable count, the degree-six test determines Lorentzian membership throughout this particular coefficient family in all degrees $N \geq 6$. This conclusion follows from the classification, not from extrapolation of a finite computation. The proof uses elementary coefficient counting, explicit congruences, and positive polynomial identities. The normalized Schur boundary point uses [2, Theorem 3].

The displayed parameter classification appears not to have been recorded in the primary sources checked. This restricted originality assessment neither asserts a general Giambelli theorem nor replaces a complete bibliographic search or independent mathematical review. The general rank-two Giambelli-immanant problem is outside the result.

10. Reproducibility

The exact companion programs check nine symbolic identities and the positive coefficient table using rational polynomial arithmetic. Independent coefficient-Hessian tests cover the exceptional profiles and the prescribed-cutoff construction. They use no floating-point inertia decisions. Their finite coverage is supplementary: the classification follows from the displayed profile reduction and identities for all stated parameters, not from testing a finite collection of degrees. Run the supplied programs with Python 3; no third-party package is required.

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